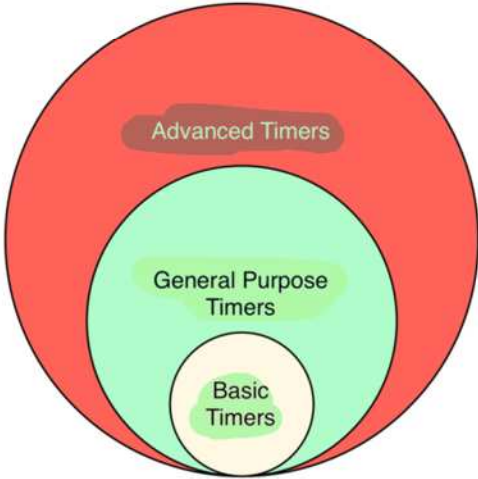


STM32 timers can mainly grouped in nine categories. Let us give a brief look at each one of them.

- **Basic timers:** timers from this category are the simplest form of timers in STM32 MCUs. They are 16-bit timers used as time base generator, and they do not have output/input pins. *Basic timers* are also used to feed the DAC peripheral, since their *update event* can trigger DMA requests for the DAC (for this reason they are usually available in STM32 MCUs providing at least a DAC). *Basic timers* can be also used as “masters” for other timers.
- **General purpose timers:** they are 16/32-bit timers (depending on the STM32-series) providing the classical features that a timer of a modern embedded microcontroller is expected to implement. They are used in any application for output compare (timing and delay generation), One-Pulse Mode, input capture (for external signal frequency measurement), sensor interface (encoder, hall sensor), etc. Obviously, a *general purpose timer* can be used as time base generator, like a *basic timer*. Timers from this category provide four-programmable input/output channels.
 - **1-channel/2-channels:** they are two subgroups of *general purpose timers* providing only one/two input/output channel.
 - **1-channel/2-channels with one complimentary output:** same as previous type, but having a *dead time* generator on one channel. This allows having complementary signals with a time base independent from the advanced timers.
- **Advanced timers:** these timers are the most complete ones in an STM32 MCU. In addition to the features found in a *general purpose timer*, they include several features related to motor control and digital power conversion applications: three complementary signals with *dead time* insertion, emergency shut-down input.
- **High-resolution timer:** The high-resolution timer (HRTIM1) is a special timer provided by some microcontrollers from the STM32F3/G4 series (which are the series dedicated to motor control and power conversion) and the STM32H7 series. It allows generating digital signals with high-accuracy timings, such as PWM or phase-shifted pulses. It consists of 6 sub-timers, 1 master and 5 slaves, totaling 10 high-resolution outputs, which can be coupled by pairs for *dead time* insertion. It also features 5 fault inputs for protection purposes and 10 inputs to handle external events such as current limitation, zero voltage or zero current switching. HRTIM1 timer is made of a digital kernel clocked at core speed followed by delay lines. Delay lines with closed loop control guarantee a 217ps resolution whatever the voltage, temperature or chip-to-chip manufacturing process deviation. The high-resolution is available on the 10 outputs in all operating modes: variable duty cycle, variable frequency, and constant ON time. This book will not cover HRTIM1 timer. ST provides a well-written Application Note, the AN4539², which cover all aspects of HRTIM timers.
- **Low-power timers:** timers from this group are especially designed for low-power applications. Thanks to their diversity of clock sources, these timers are able to keep running in *all power modes* (except for *Standby mode*). Given this capability to run even with no internal clock source, *Low-power timers* can be used as a “pulse counter” which can be useful in some applications. They also have the capability to wake up the system from *low-power* modes.

Timer Type	Counter resolution	Counter type	DMA	Channels	Complimentary channels	Synchronization	
						Master	Slave
Advanced	16-bit	up, down and center aligned	Yes	4	3	Yes	Yes
General purpose	16/32-bit	up, down and center aligned	Yes	4	0	Yes	Yes
Basic	16-bit	up	Yes	0	0	Yes	No
1-channel	16-bit	up	No	1	0	Yes (OC signal)	No
2-channels	16-bit	up	No	2	0	Yes	Yes
1-channel with one complimentary output	16-bit	up	Yes	1	1	Yes (OC signal)	No
2-channel with one complimentary output	16-bit	up	Yes	2	1	Yes	Yes
High-resolution	16-bit	up	Yes	10	10	Yes	Yes
Low-power	16-bit	up	No	1	0	No	No



Timer Type																
		STM32F0	STM32F1	STM32F3	STM32G0	STM32G4	STM32F2	STM32F4	STM32F7	STM32H7	STM32L0	STM32L1	STM32L4	STM32L4+	STM32L5	STM32U5
Advanced		TIM1	TIM1	TIM1*	TIM1	TIM1	TIM1	TIM1	TIM1	TIM1			TIM1	TIM1	TIM1	TIM1
			TIM8*	TIM8*		TIM8	TIM8	TIM8*	TIM8	TIM8			TIM8*	TIM8*	TIM8	TIM8
						TIM20*										
General purpose	16-bit	TIM3	TIM2	TIM3	TIM3	TIM3	TIM3	TIM3*	TIM3	TIM3	TIM2	TIM2	TIM3*	TIM3*	TIM3	TIM3
			TIM3	TIM4		TIM4	TIM4	TIM4*	TIM4	TIM4		TIM3	TIM4*	TIM4*	TIM4	TIM4
			TIM4*									TIM4				
			TIM5*	TIM19*												
	32-bit	TIM2		TIM2	TIM2	TIM2	TIM2	TIM2*	TIM2	TIM2			TIM2	TIM2	TIM2	TIM2
				TIM5*		TIM5	TIM5	TIM5	TIM5	TIM5		TIM5	TIM5*	TIM5*	TIM5	TIM5
Basic		TIM6*	TIM6*	TIM6	TIM6	TIM6	TIM6	TIM6*	TIM6	TIM6	TIM6	TIM6	TIM6	TIM6	TIM6	TIM6
		TIM7*	TIM7*	TIM7	TIM7	TIM7	TIM7		TIM7	TIM7	TIM7	TIM7	TIM7*		TIM7	TIM7
				TIM18*												
1-channel			TIM10*				TIM10	TIM10*	TIM10			TIM10				
			TIM11*				TIM11	TIM11	TIM11			TIM11				
			TIM13*	TIM13*			TIM13		TIM13	TIM13						
		TIM14	TIM14*	TIM14*	TIM14	TIM14	TIM14		TIM14	TIM14						
2-channels			TIM9				TIM9	TIM9	TIM9		TIM21	TIM9				
			TIM12	TIM12*			TIM12		TIM12	TIM12	TIM22*					
2-channel with one complementary output		TIM15*	TIM15*	TIM15	TIM15	TIM15				TIM15			TIM15	TIM15	TIM15	TIM15
1-channel with one complementary output		TIM16	TIM16*	TIM16	TIM16	TIM16				TIM16			TIM16	TIM16	TIM16	TIM16
		TIM17	*TIM17	TIM17	TIM17	TIM17				TIM17			TIM17*		TIM17	TIM17
High-resolution				HRTIM*		HRTIM1*										
Low-power					LPTIM1	LPTIM1		LPTIM1	LPTIM1	LPTIM1	LPTIM1		LPTIM1	LPTIM1	LPTIM1	LPTIM1
					LPTIM2	LPTIM2				LPTIM2			LPTIM2	LPTIM2	LPTIM2	LPTIM2
										LPTIM3					LPTIM3	LPTIM3
										LPTIM4						

TIM1 and TIM8 introduction

The advanced-control timers (TIM1 and TIM8) consist of a 16-bit auto-reload counter driven by a programmable prescaler.

It may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

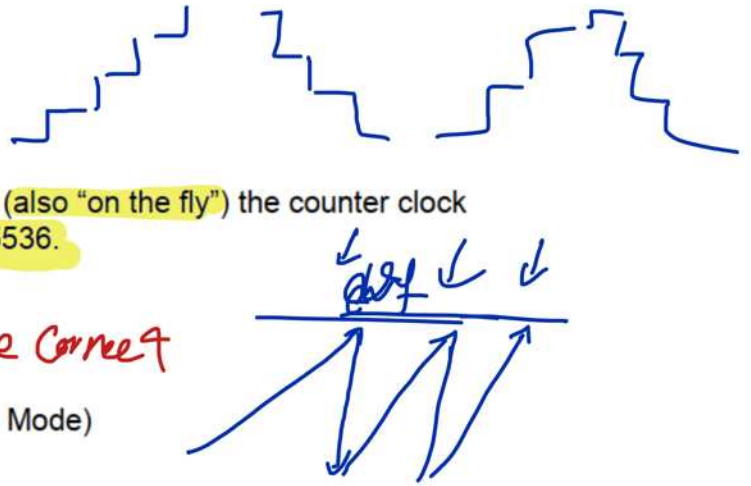
Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The advanced-control (TIM1 and TIM8) and general-purpose (TIMx) timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 14.3.20](#).

TIM1 and TIM8 main features

TIM1 and TIM8 timer features include:

- 16-bit up, down, up/down auto-reload counter.
- 16-bit programmable prescaler allowing dividing (also "on the fly") the counter clock frequency either by any factor between 1 and 65536.
- Up to 4 independent channels for:
 - Input Capture *Last*
 - Output Compare *Phase correct*
 - PWM generation (Edge and Center-aligned Mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Synchronization circuit to control the timer with external signals and to interconnect several timers together.
- Repetition counter to update the timer registers only after a given number of cycles of the counter. *8hrt = 256*
- Break input to put the timer's output signals in reset state or in a known state.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
 - Break input
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management



TIM2 to TIM5 introduction

The general-purpose timers consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (*input capture*) or generating output waveforms (*output compare and PWM*).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 15.3.15](#).

TIM9 to TIM14 introduction

The TIM9 to TIM14 general-purpose timers consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used for a variety of purposes, including measuring the pulse lengths of input signals (*input capture*) or generating output waveforms (*output compare, PWM*).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The TIM9 to TIM14 timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 16.3.12](#).

TIMx main features

General-purpose TIMx timer features include:

- 16-bit up, down, up/down auto-reload counter.
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65536.
- Up to 4 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (Edge- and Center-aligned modes)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management

TIM9/TIM12 main features

The features of the TIM9 to TIM14 general-purpose timers include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- Up to 2 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Synchronization circuit to control the timer with external signals and to interconnect several timers together
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software or internal trigger)
 - Trigger event (counter start, stop, initialization or count by internal trigger)
 - Input capture
 - Output compare

TIM10/TIM11 and TIM13/TIM14 main features

The features of general-purpose timers TIM10/TIM11 and TIM13/TIM14 include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65536 (can be changed “on the fly”)
- independent channel for:
 - Input capture
 - Output compare
 - PWM generation (edge-aligned mode)
 - One-pulse mode output
- Interrupt generation on the following events:
 - Update: counter overflow, counter initialization (by software)
 - Input capture
 - Output compare

TIM6 and TIM7 introduction

The basic timers TIM6 and TIM7 consist of a 16-bit auto-reload counter driven by a programmable prescaler.

They may be used as generic timers for time-base generation but they are also specifically used to drive the digital-to-analog converter (DAC). In fact, the timers are internally connected to the DAC and are able to drive it through their trigger outputs.

The timers are completely independent, and do not share any resources.

TIM6 and TIM7 main features

Basic timer (TIM6 and TIM7) features include:

- 16-bit auto-reload upcounter
- 16-bit programmable prescaler used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65536
- Synchronization circuit to trigger the DAC
- Interrupt/DMA generation on the update event: counter overflow

Figure 52. Advanced-control timer block diagram

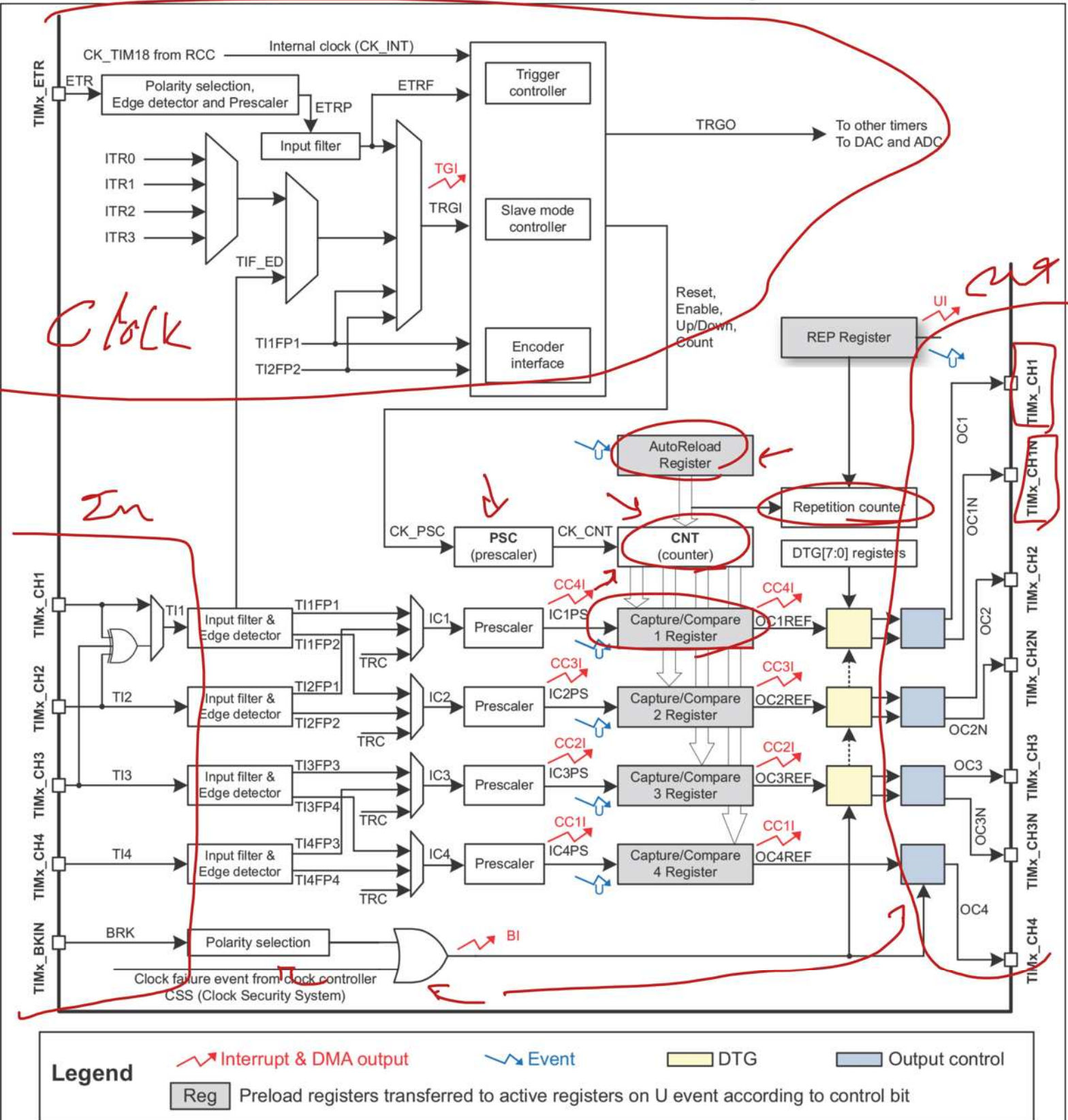
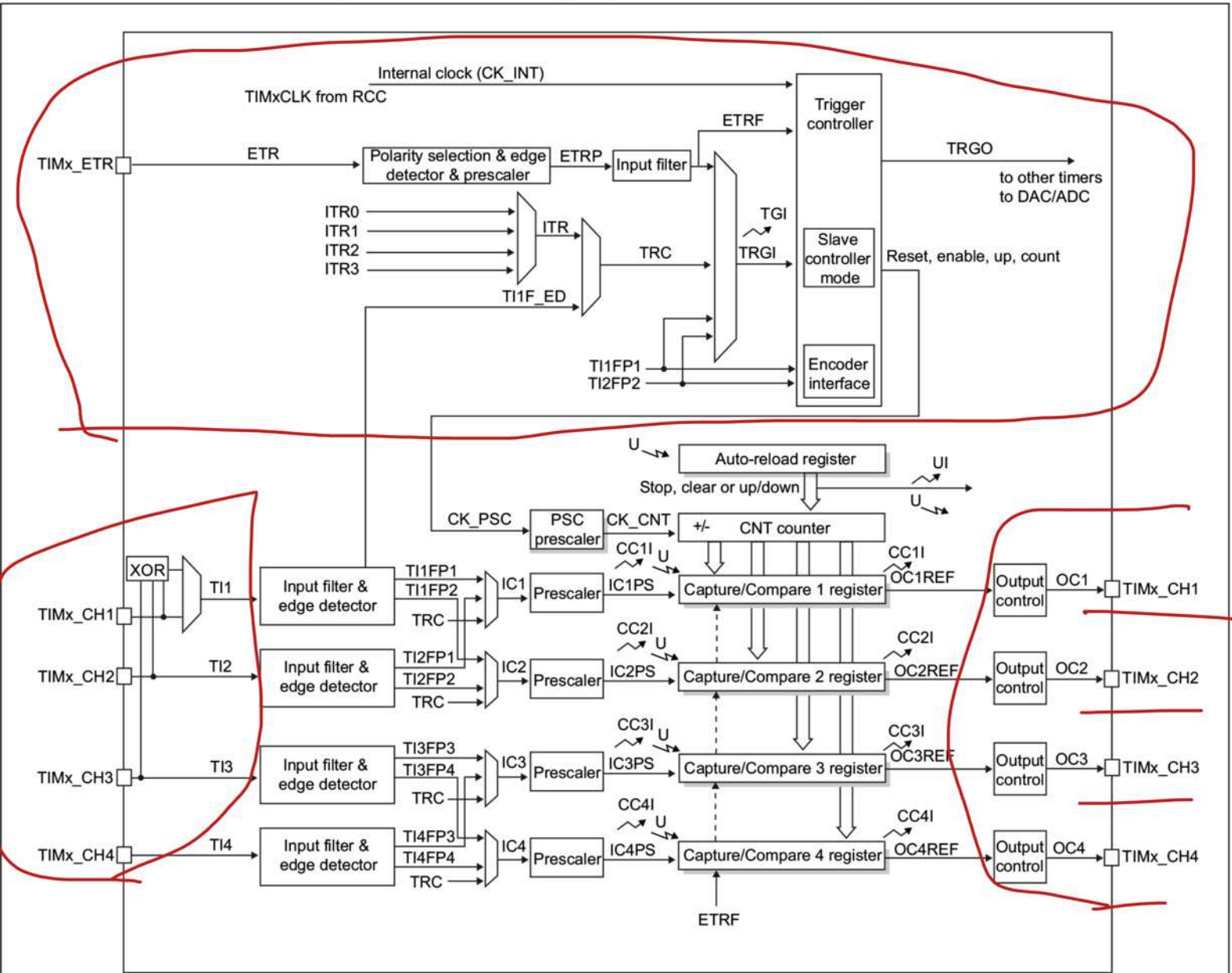


Figure 100. General-purpose timer block diagram



Notes:

Reg Preload registers transferred to active registers on U event according to control bit

⚡ Event

⚡ Interrupt & DMA output

Figure 169. Basic timer block diagram

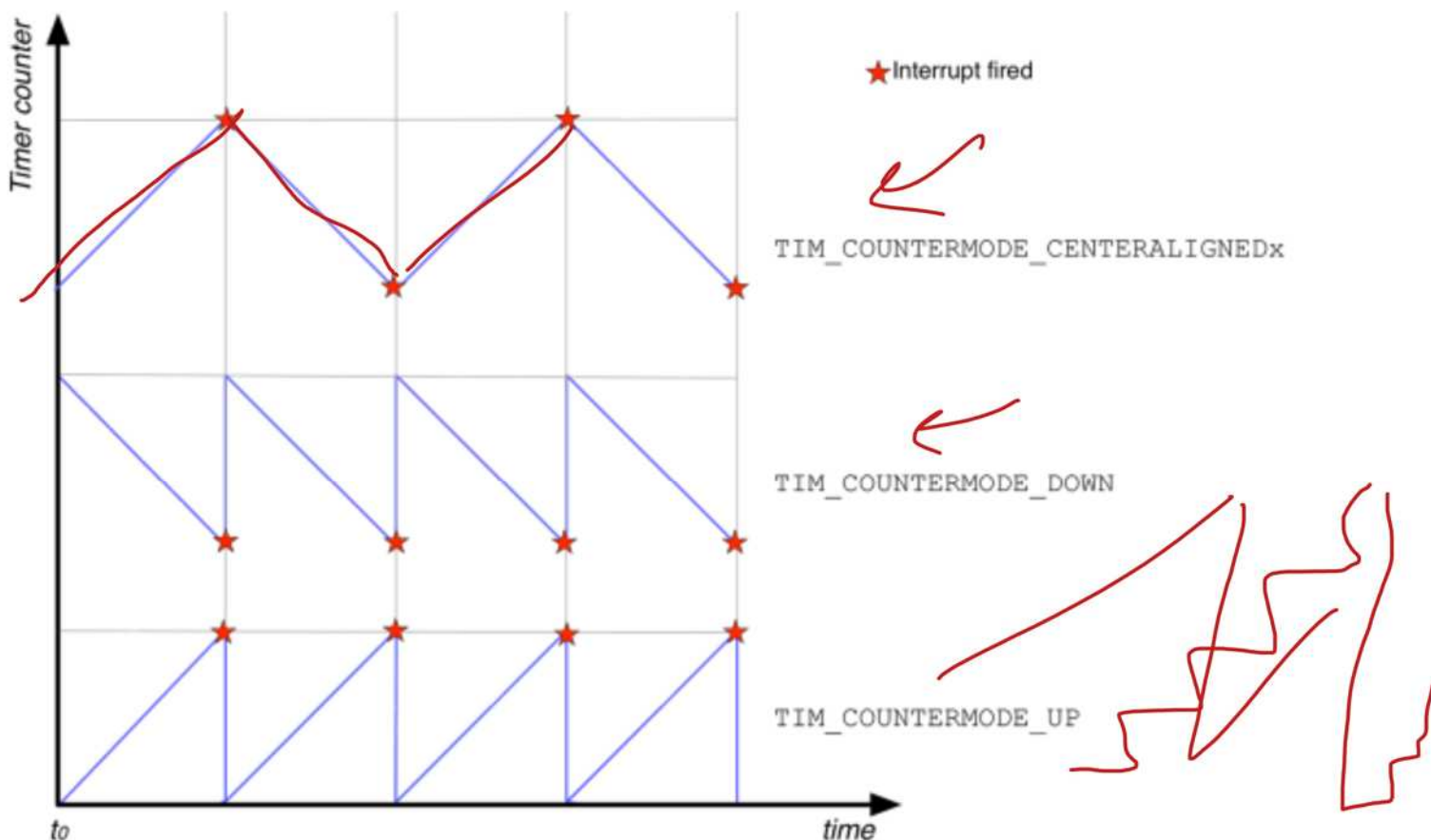
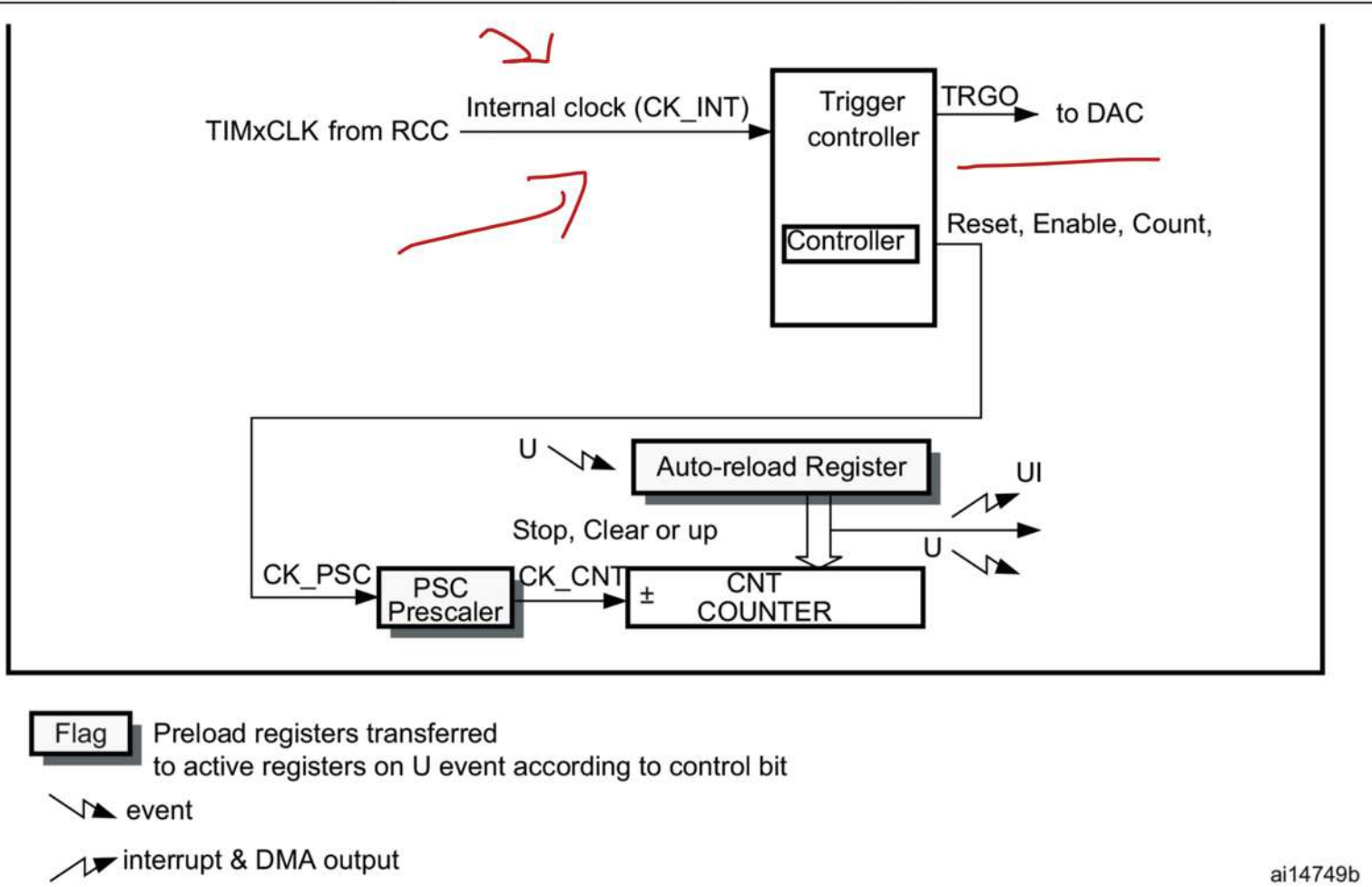
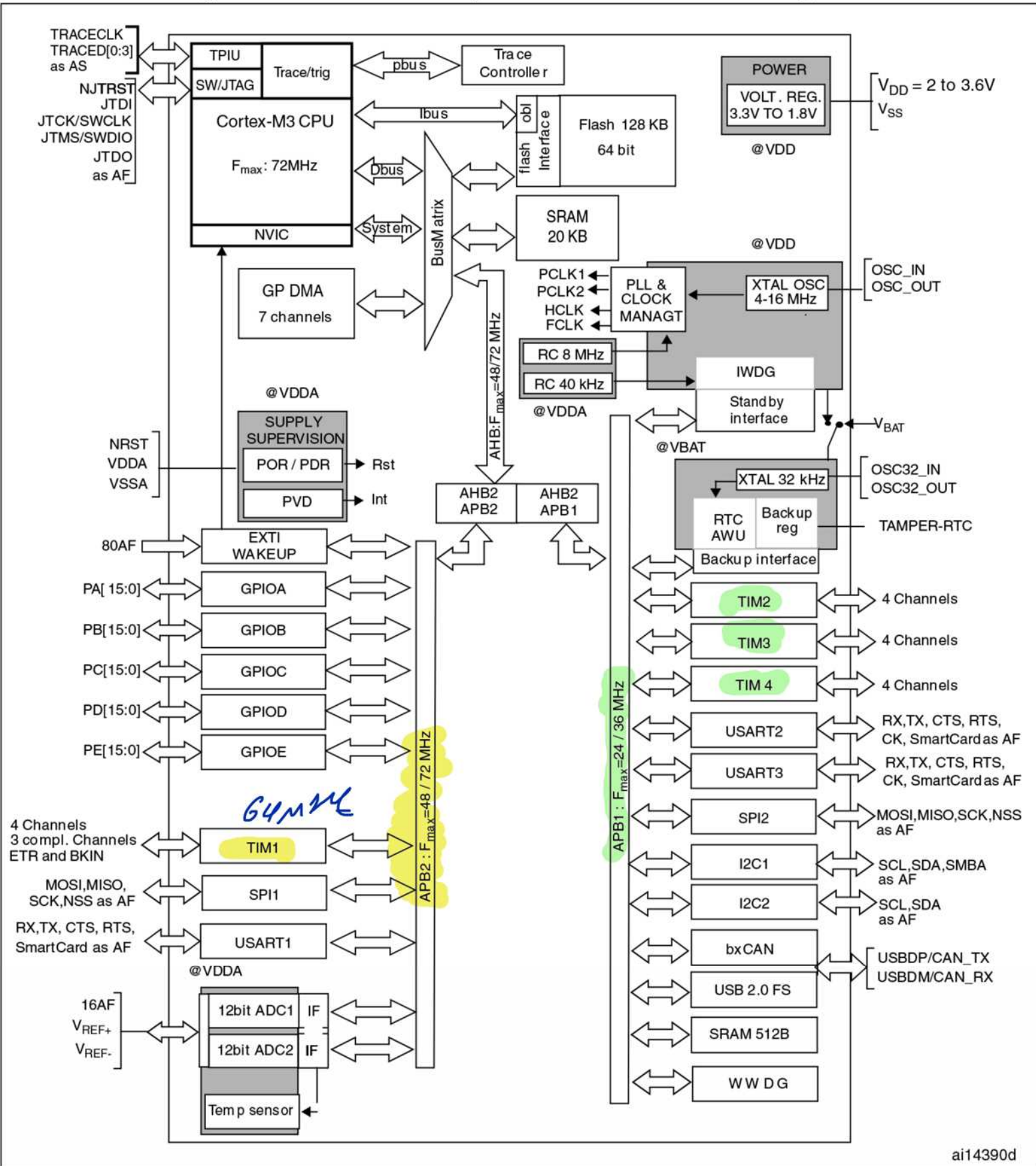


Figure 11.15: The three major counting modes of a *general purpose* timer

Figure 1. STM32F103xx performance line block diagram



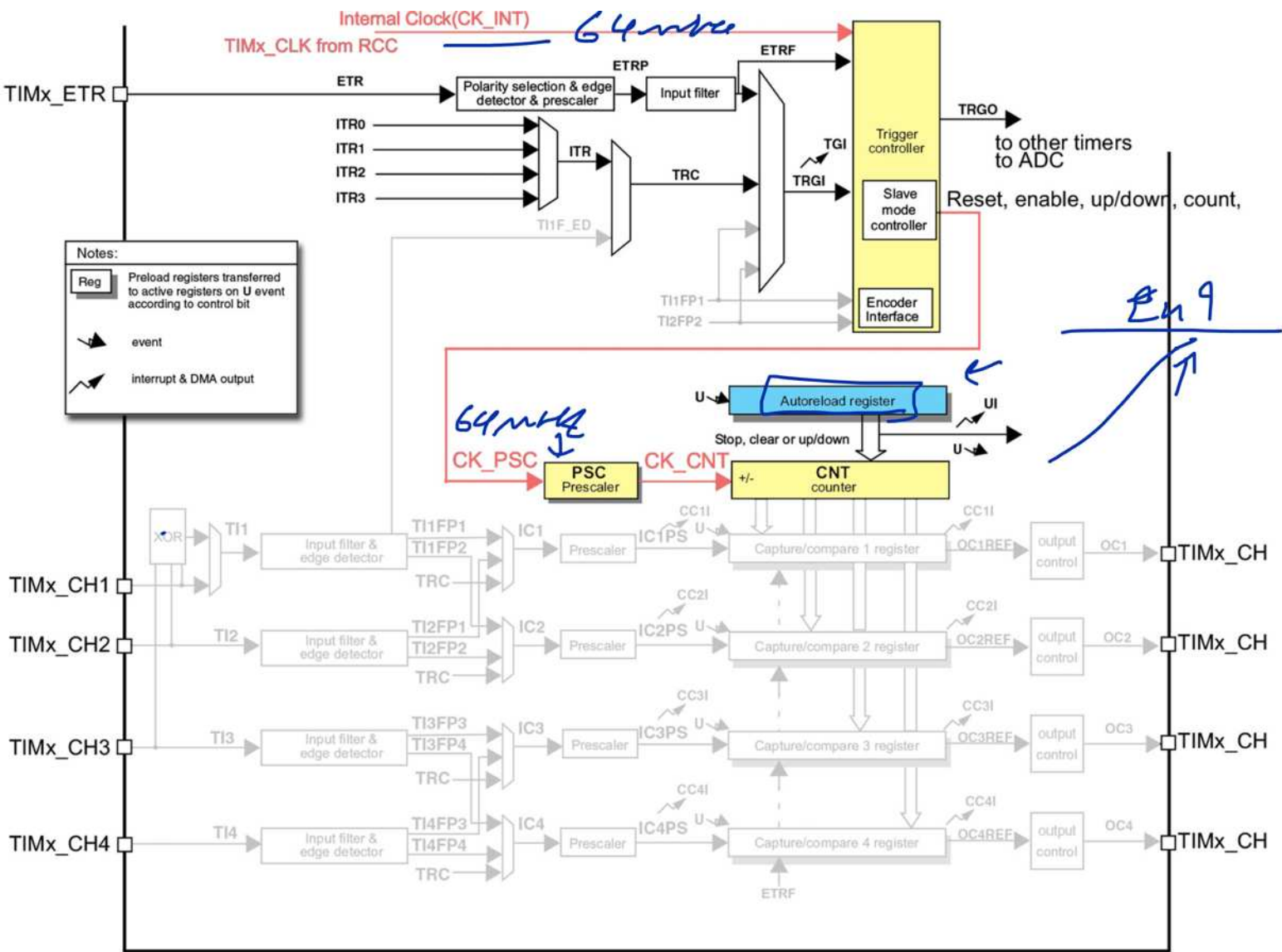


Figure 11.3: The structure of a *general purpose* timer

$$f_{\text{Timer}} = \frac{f_{\text{clk}}}{\text{prescaler}} = \frac{64 \text{ MHz}}{PR} \rightarrow 1 - 65536$$

$$\frac{1}{f} = \left(\frac{\text{prescaler}}{f_{\text{clk}}} \right) \times (ARR) \rightarrow 1 - 65536$$

$$\Rightarrow T = \frac{(PSC+1) \times (ARR+1)}{f_{\text{clk}}} \quad \begin{matrix} PSC \\ \& \\ ARR \end{matrix} \Rightarrow 0 \sim 65535$$

$$1\text{ms} \Rightarrow 1 \times 10^{-3}$$

$$f_{\text{clk}} = 64\text{MHz} \Rightarrow 64 \times 10^6$$

$$1 \times 10^{-3} = \frac{(PSC+1)(ARR+1)}{64 \times 10^6} \quad \begin{array}{l} 0 \leq ARR \leq 65535 \\ 0 \leq PSC \leq 65535 \end{array}$$

$$1 \times 10^{-3} = \frac{(PSC+1)(65535+1)}{64 \times 10^6}$$

Since $\Rightarrow$ $PSC = 0$

$$1 \times 10^{-3} = \frac{(ARR+1)}{64 \times 10^6} \Rightarrow 64 \times 10^3 = ARR+1$$

$$64 \text{ — } = ARR+1$$

$ARR = 63999$

Figure 59. Counter timing diagram, update event when ARPE=0 (TIMx_ARR not preloaded)

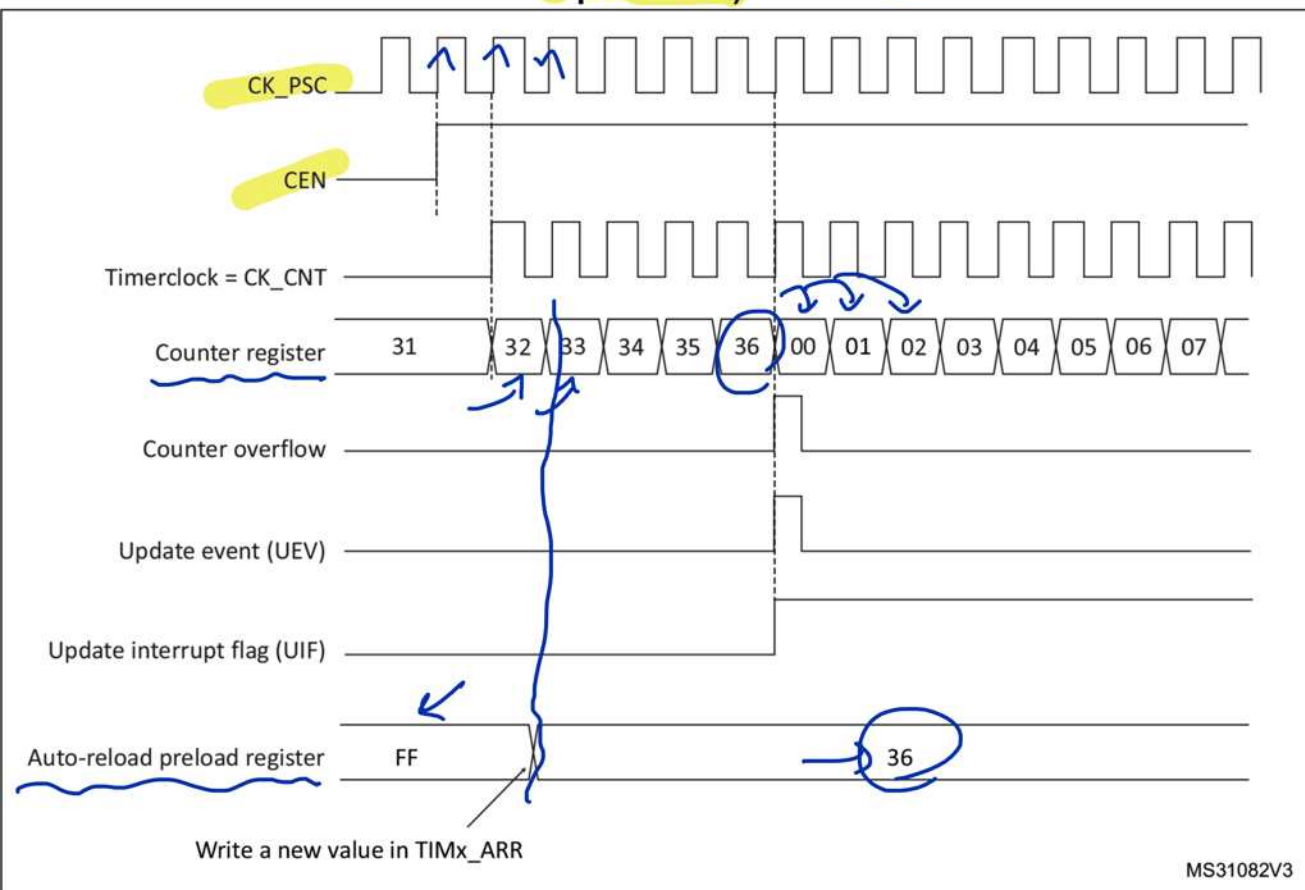


Figure 60. Counter timing diagram, update event when ARPE=1 (TIMx_ARR preloaded)

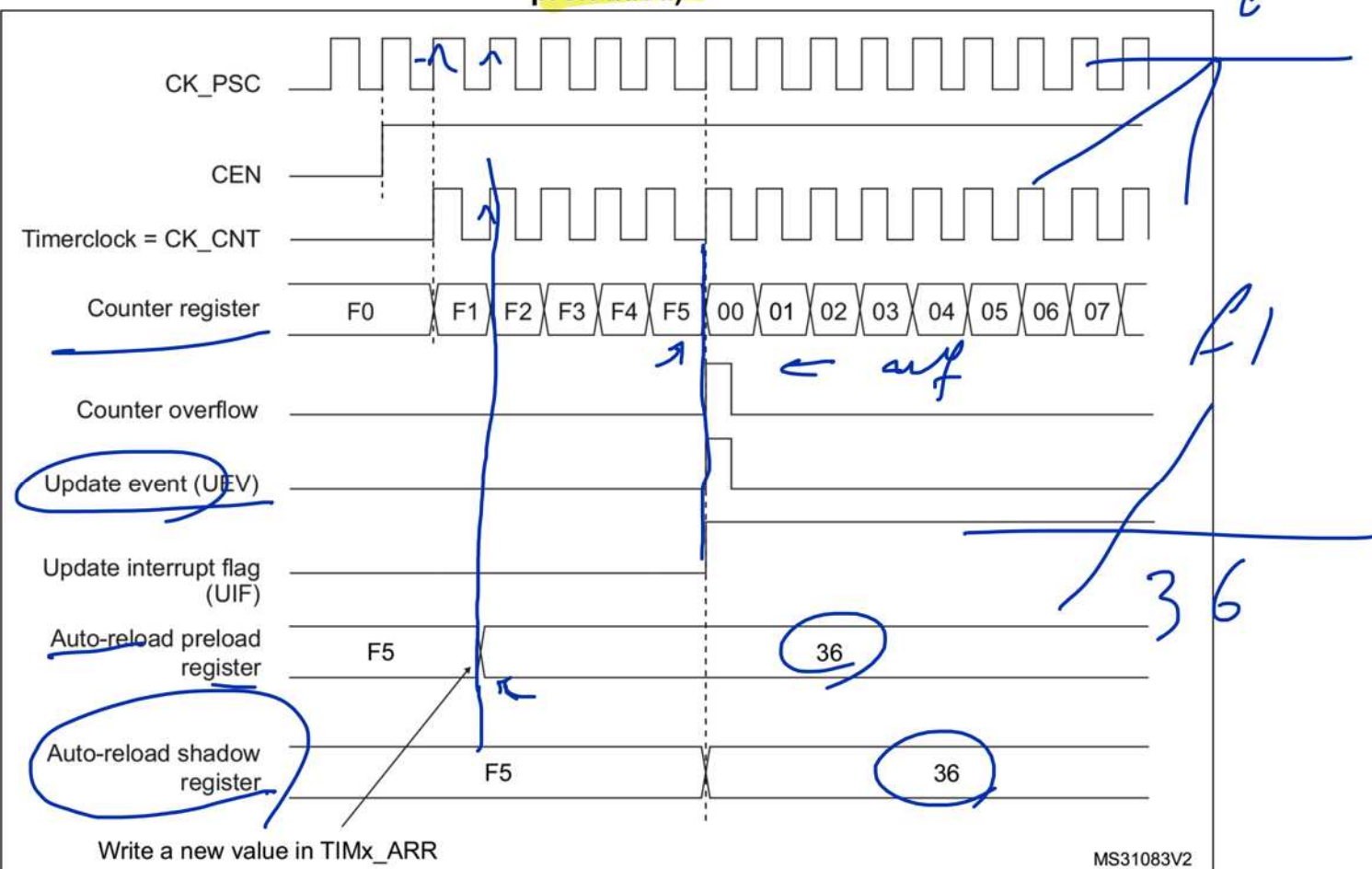


Figure 61. Counter timing diagram, internal clock divided by 1

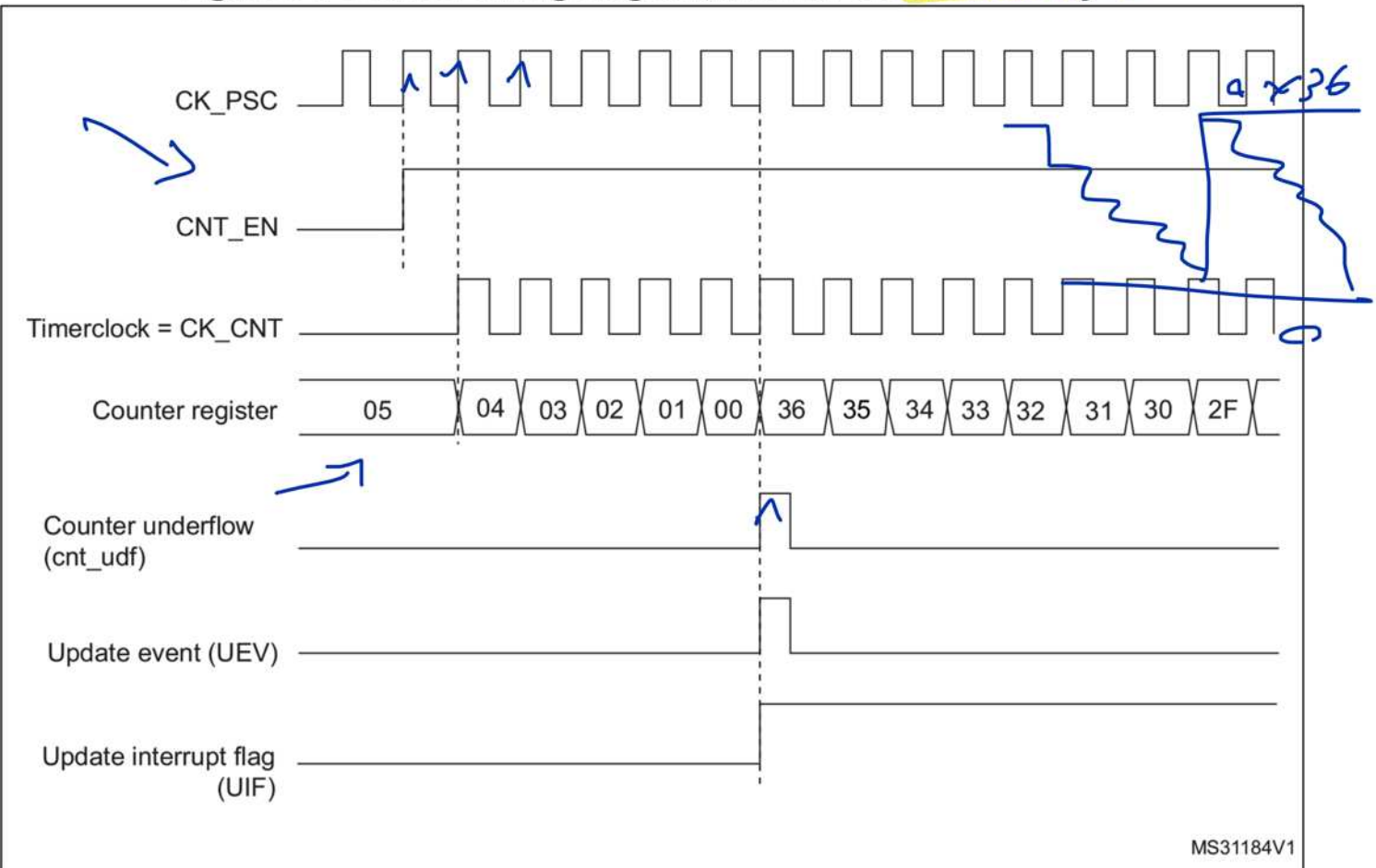


Figure 62. Counter timing diagram, internal clock divided by 2

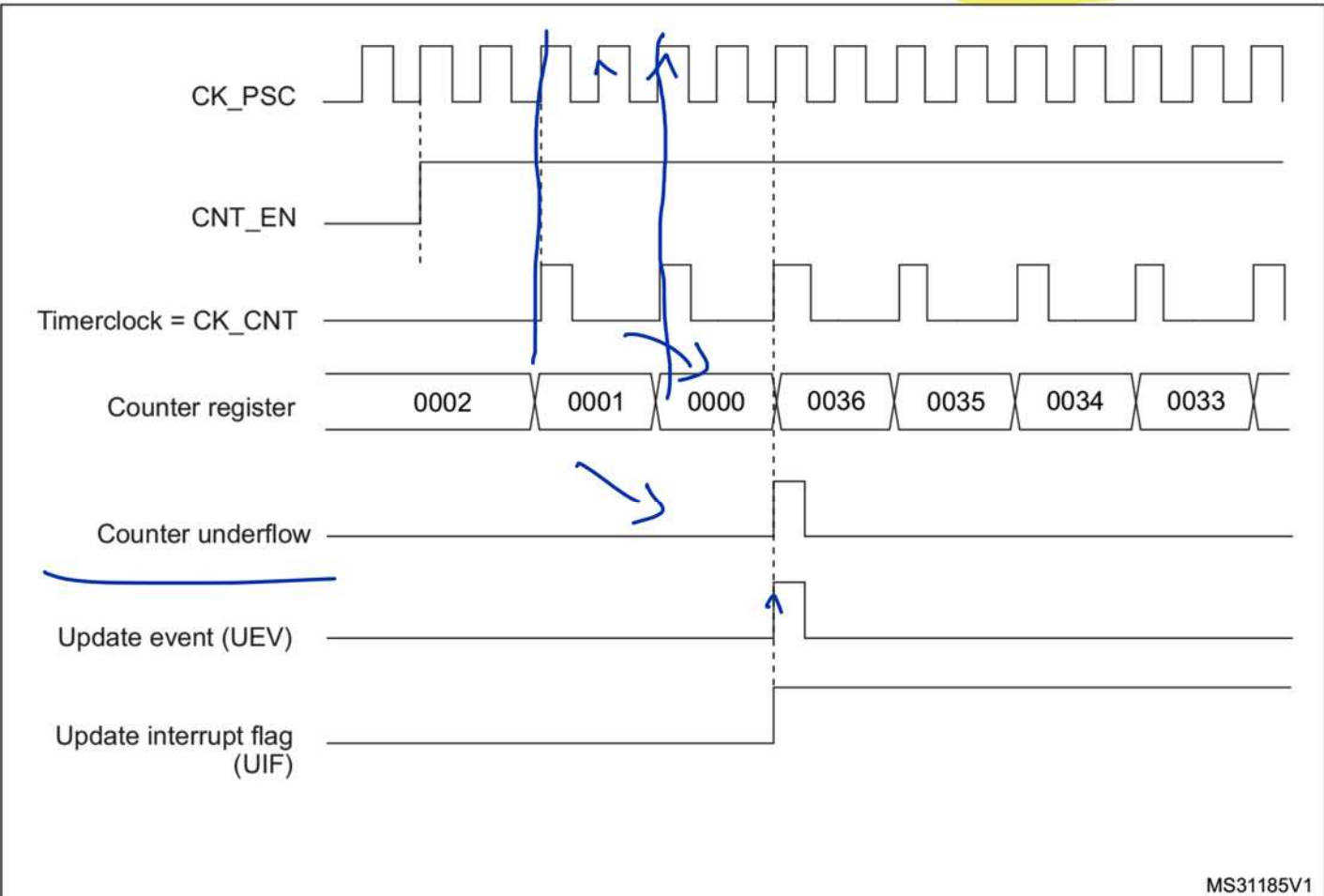
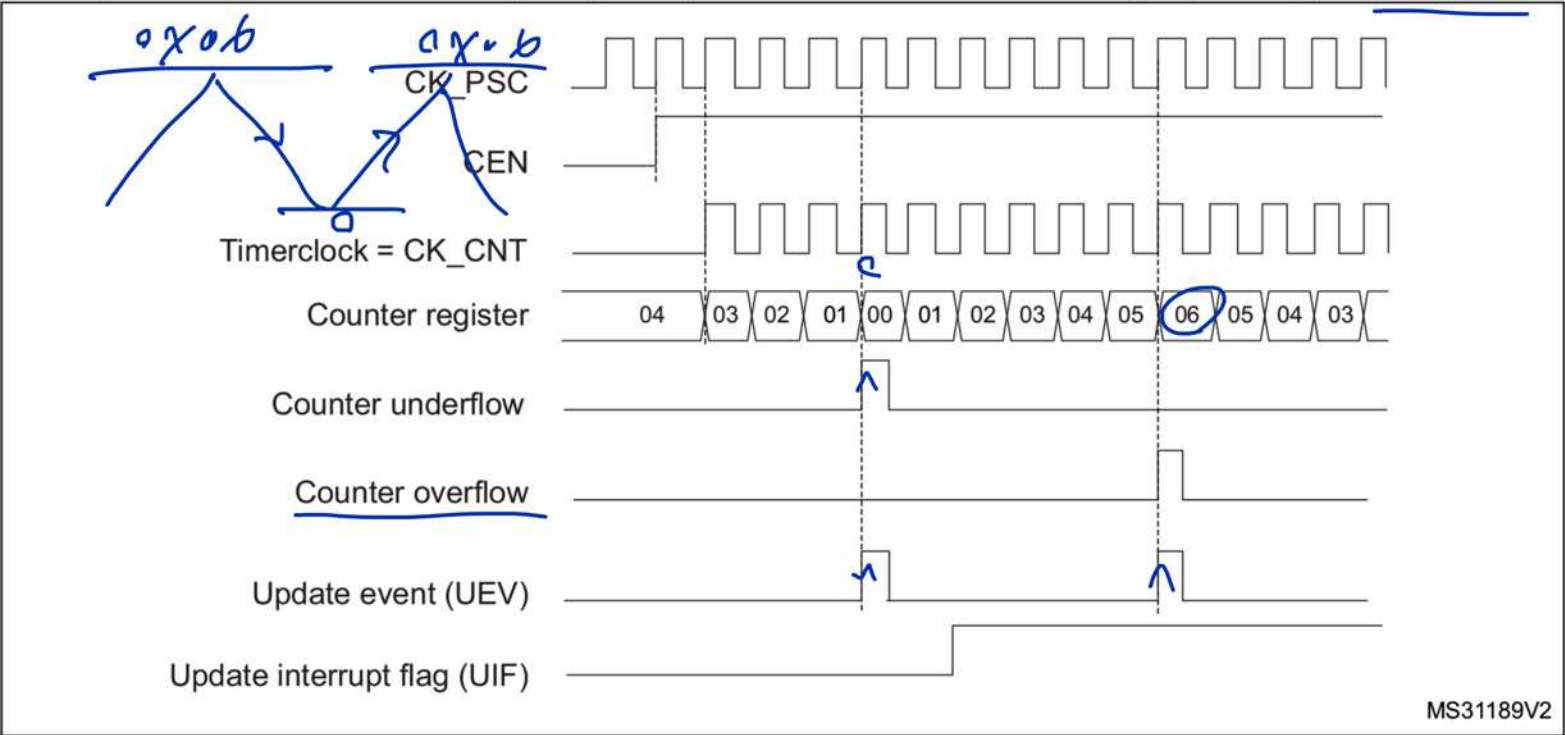


Figure 66. Counter timing diagram, internal clock divided by 1, TIMx_ARR = 0x6



MS31189V2

1. Here, center-aligned mode 1 is used (for more details refer to [Section 14.4: TIM1 and TIM8 registers](#)).

HAL_TIM_Base_Start_IT

Function name

HAL_StatusTypeDef HAL_TIM_Base_Start_IT (TIM_HandleTypeDef * htim)

Function description

Starts the TIM Base generation in interrupt mode.

Parameters

- **htim**: TIM Base handle

Return values

- **HAL**: status

HAL_TIM_PeriodElapsedCallback

Function name

void HAL_TIM_PeriodElapsedCallback (TIM_HandleTypeDef * htim)

Function description

Period elapsed callback in non-blocking mode.

Parameters

- **htim**: TIM handle

Return values

- **None**:

HAL_TIM_Base_Stop_IT

Function name

HAL_StatusTypeDef HAL_TIM_Base_Stop_IT (TIM_HandleTypeDef * htim)

Function description

Stops the TIM Base generation in interrupt mode.

Parameters

- **htim**: TIM Base handle

Return values

- **HAL**: status

__HAL_TIM_GET_IT_SOURCE

Description:

- Check whether the specified TIM interrupt source is enabled or not.

Parameters:

- **__HANDLE__**: TIM handle
- **__INTERRUPT__**: specifies the TIM interrupt source to check. This parameter can be one of the following values:
 - TIM_IT_UPDATE: Update interrupt
 - TIM_IT_CC1: Capture/Compare 1 interrupt
 - TIM_IT_CC2: Capture/Compare 2 interrupt
 - TIM_IT_CC3: Capture/Compare 3 interrupt
 - TIM_IT_CC4: Capture/Compare 4 interrupt
 - TIM_IT_COM: Commutation interrupt
 - TIM_IT_TRIGGER: Trigger interrupt
 - TIM_IT_BREAK: Break interrupt

Return value:

- The: state of TIM_IT (SET or RESET).

__HAL_TIM_SET_PRESCALER

Description:

- Set the TIM Prescaler on runtime.

Parameters:

- **__HANDLE__**: TIM handle.
- **__PRESC__**: specifies the Prescaler new value.

Return value:

- None

__HAL_TIM_SET_COUNTER

Description:

- Set the TIM Counter Register value on runtime.

Parameters:

- **__HANDLE__**: TIM handle.
- **__COUNTER__**: specifies the Counter register new value.

Return value:

- None

__HAL_TIM_GET_COUNTER

Description:

- Get the TIM Counter Register value on runtime.

Parameters:

- **__HANDLE__**: TIM handle.

Return value:

- 16-bit: or 32-bit value of the timer counter register (TIMx_CNT)

`__HAL_TIM_SET_AUTORELOAD`

Description:

- Set the TIM Autoreload Register value on runtime without calling another time any Init function.

Parameters:

- `__HANDLE__`: TIM handle.
- `__AUTORELOAD__`: specifies the Counter register new value.

Return value:

- None

`__HAL_TIM_GET_AUTORELOAD`

Description:

- Get the TIM Autoreload Register value on runtime.

Parameters:

- `__HANDLE__`: TIM handle.

Return value:

- 16-bit: or 32-bit value of the timer auto-reload register(TIMx_ARR)

`__HAL_TIM_SET_COMPARE`

Description:

- Set the TIM Capture Compare Register value on runtime without calling another time ConfigChannel function.

Parameters:

- `__HANDLE__`: TIM handle.
- `__CHANNEL__`: TIM Channels to be configured. This parameter can be one of the following values:
 - `TIM_CHANNEL_1`: TIM Channel 1 selected
 - `TIM_CHANNEL_2`: TIM Channel 2 selected
 - `TIM_CHANNEL_3`: TIM Channel 3 selected
 - `TIM_CHANNEL_4`: TIM Channel 4 selected
- `__COMPARE__`: specifies the Capture Compare register new value.

Return value:

- None

`__HAL_TIM_GET_COMPARE`

Description:

- Get the TIM Capture Compare Register value on runtime.

Parameters:

- `__HANDLE__`: TIM handle.
- `__CHANNEL__`: TIM Channel associated with the capture compare register This parameter can be one of the following values:
 - `TIM_CHANNEL_1`: get capture/compare 1 register value
 - `TIM_CHANNEL_2`: get capture/compare 2 register value
 - `TIM_CHANNEL_3`: get capture/compare 3 register value
 - `TIM_CHANNEL_4`: get capture/compare 4 register value

Return value:

- 16-bit: or 32-bit value of the capture/compare register (TIMx_CCRy)